INCH-POUND
MIL-B-18/11E
22 November 2010
SUPERSEDING
MIL-B-18/11D
21 May 1985

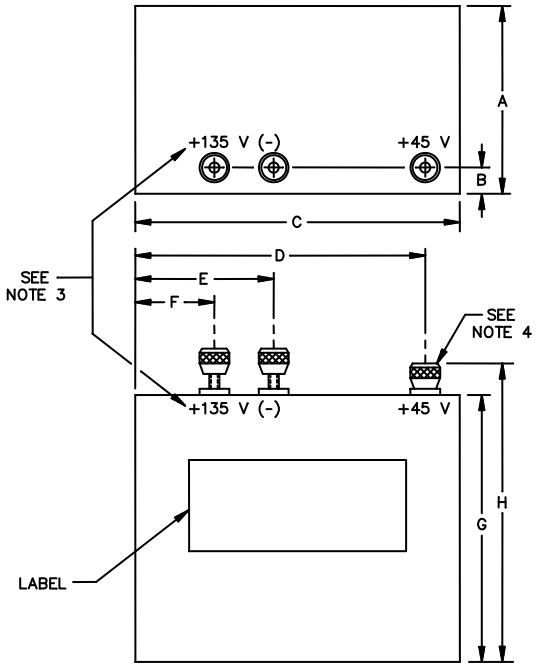
MILITARY SPECIFICATION SHEET

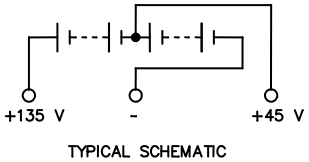
BATTERY, DRY, BA-33

INACTIVE FOR NEW DESIGN AFTER
10 AUGUST 1998 AND IS NO LONGER USED,
EXCEPT FOR REPLACEMENT PURPOSES.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-B-18.





TYPICAL SCHEMATIC

FIGURE 1. Battery dimensions.

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Ltr	Inches		mm	
	Min	Max	Min	Max
A	3.438	3.688	87.33	93.68
B	.438	.563	11.13	14.30
C	6.062	6.250	153.97	158.75
D	5.438	5.563	138.13	141.30
E	2.563	2.688	65.10	68.28
F	1.438	1.563	36.53	39.70
G	5.001	5.126	127.03	130.20
H	---	5.781	---	146.84

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Voltage and polarity markings may be placed either on the top or front face of the battery.
4. Provide insulated knurled nuts with brass, aluminum, or equivalent inserts.
5. Marking to be in accordance with MIL-B-18.

FIGURE 1. Battery dimensions - Continued.

REQUIREMENTS:

Dimensions, marking, and configuration: See figure 1.

Nominal voltage: 135 volts.

Usual number and type of cells: 90 “A” cells.

Usual cell connection: Series.

Terminals: Stud and nut.

Weight (maximum): 6 pounds, 8 ounces.

Capacity tests: When the battery is tested in accordance with the methods of examination and test of this specification, the minimum capacity-test requirements shall be not less than the minimum time specified for SLD or SLT.

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First article inspection in accordance with MIL-B-18:

Visual and mechanical (external).
Battery voltage.
Vibration test.
Mechanical-shock test in accordance with MIL-STD-202, method 213, test condition I.
Insulation resistance test in accordance with MIL-STD-202, method 302, test condition B, tolerance ±20 volts.
Capacity, D (without storage).
Jacket integrity test.

Conformance inspection in accordance with MIL-B-18:

Visual and mechanical (external).
Battery voltage.
Insulation-resistance test in accordance with MIL-STD-202, method 302, test condition B, tolerance ±20 volts.
Vibration test.
Mechanical-shock test in accordance with MIL-STD-202, method 213, test condition I.
Jacket integrity test.
Electrode leakage.
Capacity, D and T.

Method of examination and test:

Capacity tests: See requirements for capacity specified herein.

Storage:

Test	Period
D	12 months
T	90 days

Discharge: The battery shall be discharged through 20 ohms continuously to a test-end voltage of 100.0 volts.

Closed circuit voltage: Use TS-183 ( )/U test per special marking or, with minimum permissible voltage as specified for TS-183 ( )/U, use a load resistance value of 7,230 ohms.

Special marking on each unit package:

TO TEST THIS BATTERY WITH TS-183 ( )/U:

USE JACK NO.	MINIMUM PERMISSIBLE VOLTAGE
28	126.0

Unless expressly authorized, this test information shall apply to Army and Air Force applications only.

Part or Identifying Number (PIN): BA-33, see 1.2 of MIL-B-18.

Reference documents. In addition to MIL-B-18, this document references the following:

MIL-STD-202

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The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Navy - SH
Air Force - 99
DLA - CC

Preparing activity:
DLA - CC

(Project 6135-2010-047)

Review activities:
Navy - MC

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.daps.dla.mil.

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